AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

APRIL/ MAY 2019

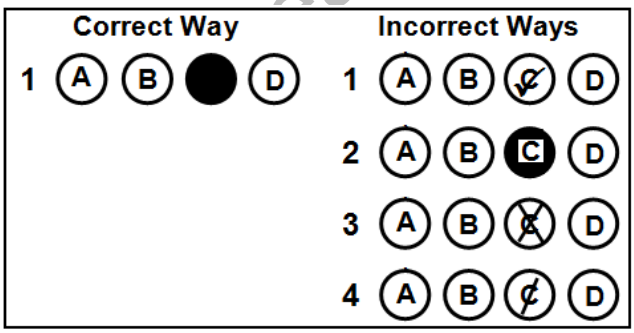
General Mathematics Paper I

Time: 45 minutes Marks: 30

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. In lowest term, the expression $\frac{4(-xy)^4}{(x^0+1)^2 y^4}$ will be

   A. $-x^4$
   B. x^4
   C. $4x^4$
   D. $-4x^4$

2. On rationalisation of $\frac{1}{\sqrt{1+a}}$, we get

   A. $\frac{\sqrt{1-a}}{1-a^2}$
   B. $\frac{\sqrt{1-a}}{(1-a)^2}$
   C. $\frac{\sqrt{1+a}}{1+a}$
   D. $\frac{\sqrt{1+a}}{1+a^2}$

3. If $a+b=-1$, then $(a+c+b)^2$ reduces to

   A. $-1-2c+c^2$
   B. $-1+c^2$
   C. $1+c^2$
   D. $1-2c+c^2$

4. On factorisation of $2^2x-2^3(x-1)$, we get

   A. $-2^2(x-2)$
   B. $-2^2(x+2)$
   C. $2^2(x-2)$
   D. $2^2(x+2)$

5. Given that the polynomial $m(-x)^3+nx^2$ is divided by $x+1$, the remainder will be

   A. $-m-n$
   B. $-m+n$
   C. $m-n$
   D. $m+n$

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6. If n is an odd integer, then the zero of polynomial $x^n - 1$

A. is 0
B. is 1
C. is -1
D. cannot be determined.

7. If P_1 and P_2 are the two polynomials and the highest common factor (HCF) is H, then the least common multiple (LCM) is

A. $\frac{H}{P_1} \times P_2$
B. $\frac{P_1}{H} \times P_2$
C. $\frac{H}{P_2} \times P_1$
D. $H \times P_1 P_2$

8. On simplification of $\frac{(p+q)(p-q)}{p^3q - pq^3}$, we get

A. $\frac{1}{pq}$
B. $\frac{1}{p-q}$
C. $\frac{p+q}{pq(p-q)}$
D. $\frac{p^2+q^2}{pq(p-q)}$

9. The least common multiple (LCM) of $(x+a)(x+b)$ and $(2x+2a)(x+c)$ is

A. $(x+a)$
B. $2(x+a)$
C. $(x+a)(x+b)(x+c)$
D. $2(x+a)(x+b)(x+c)$

10. The value of x which satisfies the linear equation $2x - 4 = -2x$ is

A. -4
B. -1
C. 0
D. 1

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11. If $3x-6=24$, then x is equal to

A. 6
B. 10
C. 15
D. 25

12. The solution set of $x-\sqrt{a}=\sqrt{a}$ is

A. $\{0\}$
B. $\{a\}$
C. $\{2\sqrt{a}\}$
D. $\{\sqrt{2a}\}$

13. The solution set of $x^2+6x+9=0$ is

A. $\{3\}$
B. $\{-3\}$
C. $\{-3, 3\}$
D. $\{0, -3\}$

14. For the equation $ax^2+bx+c=0$, the values of x will be

A. $\dfrac{b\pm\sqrt{b^2-4ac}}{2b}$
B. $\dfrac{b\pm\sqrt{4ac-b^2}}{2b}$
C. $\dfrac{-b\pm\sqrt{4ac-b^2}}{2a}$
D. $\dfrac{-b\pm\sqrt{b^2-4ac}}{2a}$

15. Let $A^t=\begin{bmatrix}-\frac{1}{2} & 3\\ a & -4\end{bmatrix}$ and $A^t+B=O$, where O is the null matrix of order 2. The matrix B is

A. $\begin{bmatrix}-\frac{1}{2} & a\\ 3 & -4\end{bmatrix}$
B. $\begin{bmatrix}\frac{1}{2} & -3\\ -a & 4\end{bmatrix}$
C. $\begin{bmatrix}-\frac{1}{2} & 3\\ a & -4\end{bmatrix}$
D. $\begin{bmatrix}\frac{1}{2} & -a\\ -3 & 4\end{bmatrix}$

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16. The adjoint of a matrix A is given by $\begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$. The matrix A would be

A. $\begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$

B. $\begin{bmatrix} -1 & 1 \\ 2 & -3 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$

D. $\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$

17. If $\frac{1}{2}|M| = a$, then the matrix M would be

A. $\begin{bmatrix} 1 & -1 \\ a & a \end{bmatrix}$

B. $\begin{bmatrix} -2 & 4 \\ 0 & a \end{bmatrix}$

C. $\begin{bmatrix} -2 & -1 \\ a & 0 \end{bmatrix}$

D. $\begin{bmatrix} 0 & 1 \\ -a & -3 \end{bmatrix}$

18. If M and N are two rectangular matrices, then $(MN)^{-1}$ exists only when the order of matrices M and N would be

(**Note**: Determinant of matrix MN is not equal to zero.)

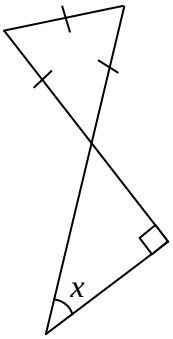
A. 3×4 and 4×1
B. 4×3 and 3×1
C. 2×1 and 1×2
D. 1×2 and 2×3

19. For the equation $a + c = (240^\circ - c)$, where a and c are supplementary angles, the value of c

A. is 60°
B. is 120°
C. is 180°
D. cannot be determined.

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20. In the given diagram, the value of x is equal to



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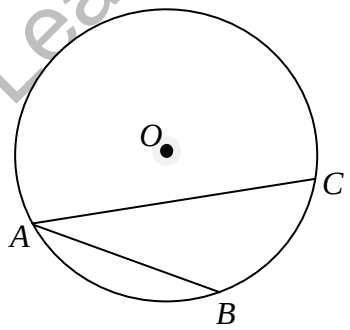
A. 30°
B. 120°
C. 150°
D. 270°

21. The INCORRECT statement(s) about a parallelogram is/are

I. diagonals of a parallelogram bisect each other and are equal
II. adjacent angles of a parallelogram are equal
III. opposite sides of a parallelogram are equal

A. I only.
B. III only.
C. I and II.
D. II and III.

22. A circle with centre O having chords AB and AC are shown in the given diagram. If $\angle AOB = m^2$, then $\angle ACB$ is equal to



A. m^2
B. $\frac{m^2}{2}$
C. $2m^2$
D. $(2m)^2$

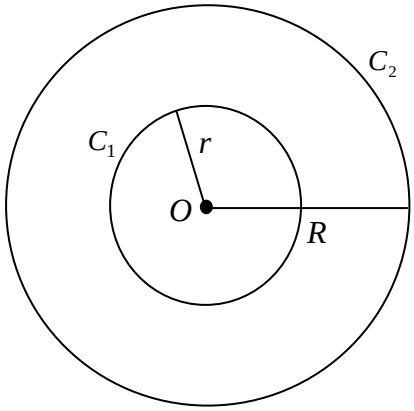
23. In a triangle, the line segment joining the midpoint of a side to the opposite vertex is called

A. angle bisector.
B. right bisector.
C. altitude.
D. median.

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Use the given information to answer Q.24 and Q.25.

The given diagram shows two concentric circles C_1 and C_2 with radii *r* and *R* respectively.



24. If the difference in areas of two concentric circles is 20 square units, then the value of $R^2 - r^2$ will be

A. $20 - \pi$ units.
B. $\pi - 20$ units.
C. $\frac{20}{\pi}$ units.
D. $\frac{\pi}{20}$ units.

25. If $r + R = 9$, then the area of circle C_2 can be expressed as

A. $\pi(9 - R)^2$
B. $\pi(9 - r)^2$
C. $\pi\left(\frac{9}{r}\right)^2$
D. $\pi\left(\frac{9}{R}\right)^2$

26. If $\frac{\text{volume}}{\text{height}} = \frac{k}{4}$ and the base area of cuboid is 12 square units, then the value of *k* is equal to

(**Formula**: Volume of cuboid = length x breadth x height)

A. 48 units.
B. 16 units.
C. 8 units.
D. 3 units.

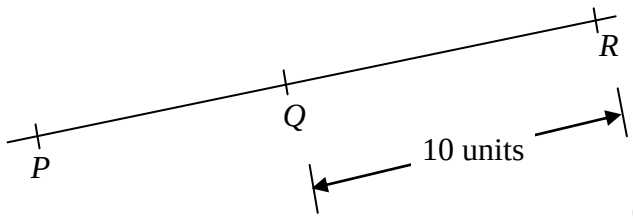
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27. If the volume of hemisphere is a cubic units, then the volume of sphere, in cubic units, will be

A. $\frac{a}{4}$
B. $\frac{a}{2}$
C. $2a$
D. $4a$

Use the given information to answer Q.28, Q.29 and Q.30.

Given that P, Q and R are three points that lie on a straight line such that $2PQ = k\,PR$ and $QR = 10$ units.



NOT TO SCALE

28. The value of $PR - PQ$ is equal to

A. 10
B. $\frac{k}{2}$
C. $2 - k$
D. $10 - \frac{k}{2}$

29. The value of $\frac{PQ}{PR}$, in terms of k, is equal to

A. $\frac{2}{k}$
B. $\frac{k}{2}$
C. $2 - k$
D. $k - 2$

30. If $PQ = QR$, then the value of k is equal to

A. 0
B. $\frac{1}{2}$
C. 1
D. 2

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